EXPERIMENT 4

Design of an Experiment about Fundamental Operations with Voltages

GROUP NO: 11.1

Objectives:

The purpose of this experiment is to analyze the behavior of signals in a circuit using multiple sinusoidal voltage sources and examine signal gain and waveforms.

Learning Outcomes:

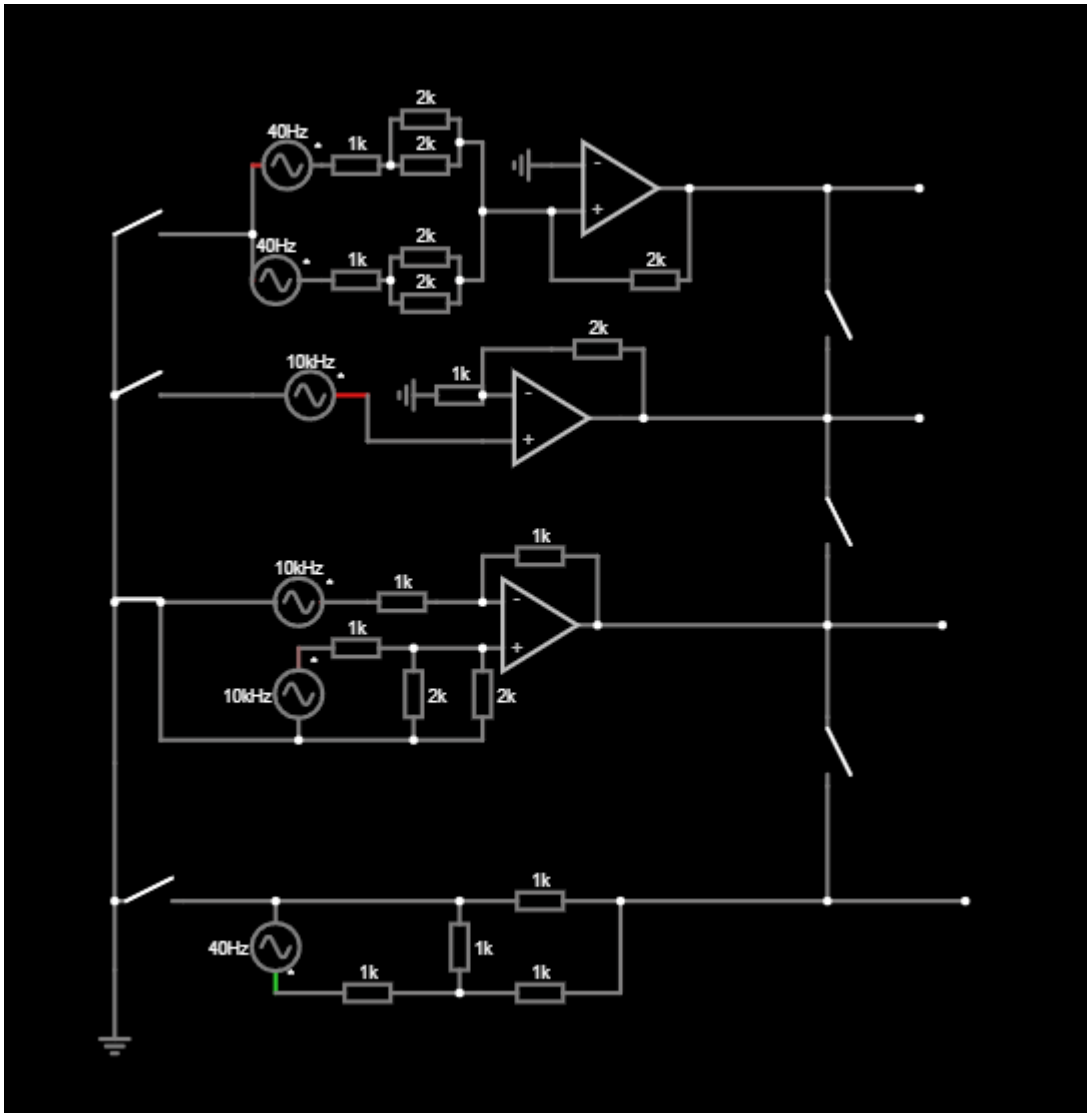
- - Understanding the basic use of operational amplifiers (Op-Amps).
- - Examining voltage divider circuits and resistive circuits.
- - Calculating signal gain and verifying measurements using an oscilloscope.

Materials:

The components and equipment required for the experiment are as follows:

- Signal Sources: V1, V2, V3, V4 ,V5 ,V6 (sinusoidal waves with different amplitudes and frequencies).
- Resistors:
- 10 x 1k Ω
- 8 x 2k Ω
- Op-Amps: U1, U2, U3
- Oscilloscope or voltmeter.

Procedure:



1. Circuit Setup:

- Using the given circuit diagram (Figure 1), place all the components correctly.

- Adjust the parameters of signal sources V1, V2, and V3:

- V1: 5 V, 40 kHz V2: 5 V, 40 kHz V3: 5 V, 10 kHz V4: 5 V 10 kHz V5: 1 V 10kHz V6: 5 V 40kHz

2. Circuit Operation:

- Measure the input and output signals of the Op-Amps using an oscilloscope or a multimeter.

- Analyze the voltage gains and waveforms.

3. Recording Data:

- Record the values at output voltage measurement points (S1, S2, S3).

- Calculate the voltage drops across resistors.

4. Analysis:

- Verify the theoretical and practical results for the Op-Amps' input/output relationships.

Question 1:
Simplify the resistances in the top most circuit
Combine the resistors in the first circuit to simplify the overall resistance using series and parallel resistance formulas.

Question 2:
Find the power of R_f in the second circuit from the top
Calculate the power dissipated by the feedback resistor (R_f) in the second circuit by finding the current and voltage across it.

Question 3:
Improve the efficiency of the third circuit
Analyze the third circuit and propose modifications to make it more efficient in terms of performance or component usage.

Question 4:
Perform a node voltage analysis on the bottom circuit
Apply the node voltage method to the last circuit to calculate the voltages at each node.

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